

BAULKHAM HILLS HIGH SCHOOL

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

2010

MATHEMATICS

EXTENSION 2

GENERAL INSTRUCTIONS:

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Start each question on a new page.
- Write your Student Number at the top of each page.
- Calculators may be used.
- A table of standard integrals is provided.
- **ALL** necessary working should be shown in every question.

QUESTION 1 (15 marks)

Marks

(a) Find each of the following integrals

$$(i) \int x^2(1 + 2x^3)^{-5} dx \quad 2$$

$$(ii) \int \tan^4 x dx \quad 3$$

$$(iii) \int \frac{dx}{3+2 \cos x} \quad 4$$

$$(b) \quad \text{Find } \int \frac{e^x + e^{2x}}{1 + e^{2x}} dx \quad 3$$

$$(c) \quad (i) \text{ Show that } \sin(A + B) + \sin(A - B) = 2 \sin A \cos B \quad 1$$

$$(ii) \text{ Hence calculate } \int \sin 5x \cos 4x dx \quad 2$$

QUESTION 2 (15 marks)

(a) For $z_1 = 2 - 3i$ and $z_2 = 1 + 5i$ find, in the form $a+ib$, the values of

$$(i) \quad z_1 + \bar{z}_2 \quad 2$$

$$(ii) \quad z_1 z_2 \quad 2$$

$$(iii) \quad \frac{z_1}{z_2} \quad 2$$

$$(b) \quad (i) \quad \text{Solve } (x + iy)^2 = 6i \quad 2$$

$$(ii) \quad \text{Hence or otherwise solve } z^2 - (1 - i)z - 2i = 0 \quad 3$$

$$(c) \quad (i) \quad \text{Express } z = 1 - \sqrt{3}i \text{ in modulus-argument form} \quad 2$$

$$(ii) \quad \text{Hence express } z^6 \text{ in the form } a + ib \quad 2$$

QUESTION 3 (15 marks)

Marks

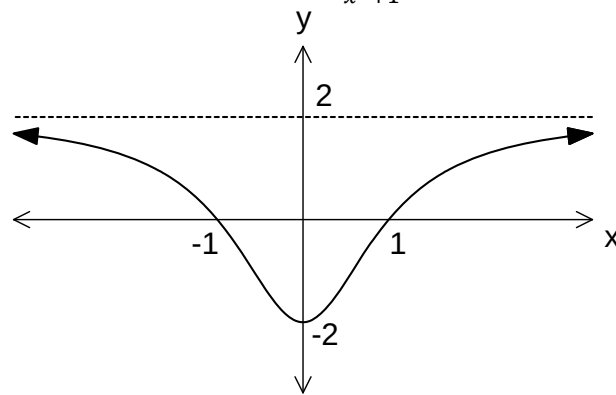
- (a) The hyperbola H has the equation $\frac{x^2}{4} - \frac{y^2}{12} = 1$
- Find
- (i) its eccentricity 2
 - (ii) the coordinates of its foci 1
 - (iii) the equations of its directrices 1
 - (iv) the equations of its asymptotes 1
- (b) (i) Show that the gradient of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ is $\frac{-b \cos \theta}{a \sin \theta}$ 2
- (ii) Hence show that the equation of the tangent is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ 3
- (iii) Show that the x-intercept of this tangent is $(\frac{a}{\cos \theta}, 0)$ 1
- (iv) Hence, or otherwise, find the points on the curve $4x^2 + 3y^2 = 12$ whose tangent passes through (2,0) 4

QUESTION 4 (15 marks)

- (a) OABC is a square on the Argand diagram and is labeled in an anticlockwise direction. A represents $z = a + ib$ and B represents $4 + 7i$.
- (i) Find, in terms of a and b , the complex number represented by C. 2
 - (ii) Hence evaluate a and b . 2
- (b) The equation $x^3 + 2x - 1 = 0$ has roots α, β and γ . Find the equation with roots:
- (i) $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ 2
 - (ii) α^2, β^2 and γ^2 2
- (c) (i) Show that 2 is a root of multiplicity 3 for $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$ 2
- (ii) Hence solve $P(x) = 0$ 2
- (d) Draw on separate argand diagrams the following loci:
- (i) $z\bar{z} = 3$ 1
 - (ii) $\arg\left(\frac{z}{z-1}\right) = \frac{\pi}{3}$ 2

QUESTION 5 (15 marks)

Marks

(a) Suppose $x > 0, y > 0, z > 0$ (i) Prove $x^2 + y^2 + z^2 \geq xy + yz + xz$ 2(ii) Given $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - xz)$
prove $x^3 + y^3 + z^3 \geq 3xyz$ 1(iii) Hence show $a + b + c \geq 3(abc)^{\frac{1}{3}}$ 1(b) Given below is the graph of $f(x) = 2 - \frac{4}{x^2+1}$.Use the graph of $y = f(x)$ to sketch, on separate axes, the graphs of(i) $y = |f(x)|$ 2(ii) $y = [f(x)]^2$ 2(iii) $y^2 = f(x)$ 2(iv) $y = \frac{1}{f(x)}$ 2(c) For the curve $x^3 + 3x^2y - 2y^3 = 16$ (i) Show that $\frac{dy}{dx} = \frac{x^2+2xy}{2y^2-x^2}$ 1

(ii) Find the coordinates of the stationary points on the curve 2

QUESTION 6 (15 marks)

- (a) Find, using slices, the volume generated when the area bounded by $y = x^2$ and the line $y = 3$ is rotated about the line $y = 3$. 4
- (b) Find, using cylindrical shells, the volume obtained by revolving about the y-axis the region bounded by the curve $y = \sin x$, for $0 \leq x \leq \pi$, and the x-axis. 4
- (c) A solid has a semi-circular base whose equation is $y = \sqrt{4 - x^2}$. Vertical cross-sections, perpendicular to the diameter, are right-angled triangles whose height is bounded by the parabola $z = 4 - x^2$.
- (i) Draw a neat diagram, including a typical slice, representing this information. 1
- (ii) By slicing at right angles to the x-axis, show that the volume of the solid is given by $\int_0^2 (4 - x^2)^{\frac{3}{2}} dx$. 3
- (iii) Hence calculate this volume. 3

QUESTION 7 (15 marks)

- (a) Use the method of partial fractions to show that $\int_0^1 \frac{6x+4}{(x^2+1)(x+1)} dx = \frac{5\pi}{4} - \frac{1}{2} \log_e 2$ Marks 4
- (b) Let $P(z) = z^4 + bz^2 + d$ where b and d are real numbers and $d \neq 0$. $P(z)$ has a double zero α .
- (i) Prove $P'(z)$ is odd. 2
- (ii) Prove that $-\alpha$ is also a double zero of $P(z)$. 2
- (c) A mass of 35 kg is dropped from a balloon falling at 30 m/s. The mass experiences air resistance measuring $70v$ Newtons, where v m/s is its velocity. Take g as 10 m/s^2 .
- (i) Show that the velocity of the mass t seconds after being dropped, but before hitting the ground, is given by $v = 5 + 25e^{-2t}$. 3
- (ii) Describe what happens to the velocity as $t \rightarrow \infty$. 1
- (iii) If the mass was dropped from 400m above the ground, how close to the ground will it be after 1 minute? 3

QUESTION 8 (15 marks)

- (a) Given $I_n = \int_0^1 x^n e^{-x} dx$
- (i) Calculate I_0 2
- (ii) Prove $I_n = nI_{n-1} - \frac{1}{e}$ 2
- (iii) Hence find $\int_0^1 x^3 e^{-x} dx$ 2
- (b) Two particles of equal mass are attached to the ends A and B of a light inextensible string which passes through a small hole at the apex C of a hollow right circular cone fixed with its axis vertical and apex on top. The semi-vertical angle of the cone is θ . The particle at A, where AC is a units, moves in a horizontal circle with constant angular velocity ω on the smooth surface of the cone, while the other particle at B hangs at rest inside the cone.
- (i) Represent this information on a diagram showing relevant forces. 1
- (ii) Show that $\omega^2 = \frac{g}{a(1+\cos\theta)}$ 2
- (iii) Hence, or otherwise, deduce that $\frac{g}{2\omega^2} < a < \frac{g}{\omega^2}$ 2
- (c) If $x > 0$, prove $x - \frac{1}{3}x^3 < \tan^{-1} x < x - \frac{1}{3}x^3 + \frac{1}{3}x^5$ 4

END OF EXAM

$$\begin{aligned}
 \text{Q1(a)(i)} \quad I &= \int x^2(1+2x^3)^{-5} dx \\
 \text{let } u &= 1+2x^3 \therefore \frac{du}{dx} = 6x^2 \\
 \therefore \frac{du}{6} &= x^2 dx \\
 \therefore I &= \frac{1}{6} \int u^{-5} du \\
 &= \frac{1}{6} \frac{u^{-4}}{-4} + C \\
 &= -\frac{1}{24} (1+2x^3)^{-4} + C \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad I &= \int \tan^4 x dx \\
 &= \int \tan^2 x \tan^2 x dx \\
 &= \int \tan^2 x (\sec^2 x - 1) dx \\
 &= \int \tan^2 x \sec^2 x - \tan^2 x dx \\
 &= \frac{1}{3} \tan^3 x - \int \sec^2 x - 1 dx \\
 &= \frac{1}{3} \tan^3 x - \tan x + x + C \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad I &= \int \frac{dx}{3+2\cos x} \\
 &= \int \frac{2dt}{\frac{1+t^2}{3+2\left(\frac{1-t^2}{1+t^2}\right)}} \\
 &= \int \frac{2dt}{3+3t^2+2-2t^2} \\
 &= \int \frac{2dt}{5+t^2} \\
 &= \frac{2}{\sqrt{5}} \tan^{-1}\left(\frac{t}{\sqrt{5}}\right) + C \\
 &= \frac{2}{\sqrt{5}} \tan^{-1}\left(\frac{1}{\sqrt{5}} \tan \frac{x}{2}\right) + C \quad (4)
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad I &= \int \frac{e^x}{1+e^{2x}} dx + \frac{1}{2} \int \frac{2e^{2x}}{1+e^{2x}} dx \\
 &= I_1 + \frac{1}{2} \ln(1+e^{2x}) + C \\
 \text{In } I_1, \text{ let } u &= e^x \therefore du = e^x dx \\
 \therefore I &= \int \frac{du}{1+u^2} + \frac{1}{2} \ln(1+e^{2x}) + C \\
 &= \tan^{-1} u + \frac{1}{2} \ln(1+e^{2x}) + C \\
 &= \tan^{-1} e^x + \frac{1}{2} \ln(1+e^{2x}) + C \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B \\
 &= 2 \sin A \cos B = \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 I &= \frac{1}{2} \int 2 \sin 5x \cos 4x dx \\
 &= \frac{1}{2} \int \sin 9x + \sin x dx \\
 &= \frac{1}{2} \left(-\frac{1}{9} \cos 9x - \cos x \right) + C \quad (3)
 \end{aligned}$$

$$\begin{aligned}
 \text{Q2(a)(i)} \quad z_1 + z_2 &= 2-3i + 1-5i \\
 &= 3-8i \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad z_1 z_2 &= (2-3i)(1-5i) \\
 &= 2+10i-3i+15 \\
 &= 17+7i \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \frac{z_1}{z_2} &= \frac{2-3i}{1-5i} \times \frac{1-5i}{1-5i} \\
 &= \frac{2-10i-3i+15}{1+25} \\
 &= \frac{-13-13i}{26} \\
 &= -\frac{1}{2} - \frac{1}{2}i \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 \text{b) (i)} \quad x^2 - y^2 + 2ixy &= 6i \\
 \therefore x^2 - y^2 &= 0 \text{ and } 2xy = 6 \\
 \therefore x^2 - \frac{9}{x^2} &= 0 \quad \therefore y = \frac{3}{x} \\
 x^4 - 9 &= 0 \\
 x^2 &= \pm 3 \text{ but } x^2 > 0 \\
 \therefore x^2 &= 3 \\
 x &= \pm \sqrt{3}, y = \pm \sqrt{3}
 \end{aligned}$$

$$\therefore z = \pm (\sqrt{3} + \sqrt{3}i) \quad (2)$$

$$\begin{aligned}
 \text{(ii)} \quad z &= \frac{(1-i) \pm \sqrt{(1-i)^2 - 4 \times 1 \times -2i}}{2 \times 1} \\
 &= \frac{1-i \pm \sqrt{1-2i-1+8i}}{2} \\
 &= \frac{1-i \pm \sqrt{6i}}{2} \\
 &= \frac{1-i \pm (\sqrt{3} + \sqrt{3}i)}{2} \\
 &= \frac{1+\sqrt{3} + (\sqrt{3}-1)i}{2}, \frac{1-\sqrt{3} + (-\sqrt{3}-1)i}{2} \quad (3)
 \end{aligned}$$

$$\text{c) (i)} \quad z = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) \quad (2)$$

$$\begin{aligned}
 \text{(ii)} \quad z^6 &= \left[2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\right]^6 \\
 &= 2^6 (\cos 2\pi + i \sin 2\pi) \\
 &= 64(1+0i) \\
 &= 64 \quad (2)
 \end{aligned}$$

Q3(a)(i) $a=2, b=\sqrt{2}=2\sqrt{3}$

15 $b^2 = a^2(e^2 - 1) \quad -1$

$12 = 4(e^2 - 1)$

$e^2 - 1 = 3$

$e^2 = 4 \quad e = 2 \quad (2)$

(ii) Foci $= (\pm ae, 0) = (\pm 4, 0) \quad (1)$

(iii) Dir. are $x = \pm \frac{a}{e} \therefore x = \pm 1 \quad (1)$

(iv) Asym. are $y = \pm \frac{bx}{a}$
 $\therefore y = \pm \frac{2\sqrt{3}x}{2} \therefore y = \pm \sqrt{3}x \quad (1)$

(b) (i) Diff. Impl.

$\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$

$\frac{2yy'}{b^2} = -\frac{2x}{a^2}$

$y' = -\frac{b^2 x}{a^2 y} \quad \checkmark$

$\therefore m_{\text{at } \theta} = -\frac{b^2 a \cos \theta}{a^2 b \sin \theta} = -\frac{b \cos \theta}{a \sin \theta} \quad (2)$

(ii) Tangent is

$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$ay \sin \theta - ab \sin^2 \theta = -bx \cos \theta + ab \cos^2 \theta$

$bx \cos \theta + ay \sin \theta = ab(\sin^2 \theta + \cos^2 \theta)$

$\frac{bx \cos \theta}{ab} + \frac{ay \sin \theta}{ab} = \frac{ab}{ab} \quad -1$

$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (3)$

(iii) x-int when $y = 0$

$\therefore \frac{x \cos \theta}{a} = 1$

$x = \frac{a}{\cos \theta}$

$\therefore x\text{-int is } \left(\frac{a}{\cos \theta}, 0\right) \quad (1)$

(iv) $4x^2 + 3y^2 = 12$

$\frac{x^2}{3} + \frac{y^2}{4} = 1$

$\therefore a = \sqrt{3}, b = 2 \quad -1$

since $a \cos \theta = 2, \cos \theta = \frac{2}{\sqrt{3}}$

$\therefore \theta = \frac{\pi}{6} \text{ or } -\frac{\pi}{6} \text{ in } Q_1 \text{ and } Q_4$

$\therefore$ Points of contact are:

$P(a \cos \theta, b \sin \theta)$

$\therefore P_1 = \left(\sqrt{3} \cos \frac{\pi}{6}, 2 \sin \frac{\pi}{6}\right) = \left(\frac{3}{2}, 1\right) \quad -1$

$P_2 = \left(\sqrt{3} \cos -\frac{\pi}{6}, 2 \sin -\frac{\pi}{6}\right) = \left(\frac{3}{2}, -1\right) \quad -1$

(4)

Q4(a)(i) $w_c = (a+ib) \times i = -1$

$= -b+ai \quad -1$

(ii) $-b+ai + a+ib = 4+7i$

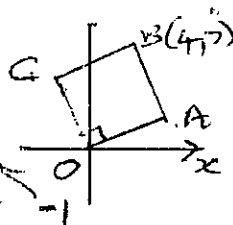
$(a-b) + (b+a)i = 4+7i \quad -1$

$\therefore a-b=4$

$b+a=7$

$\therefore 2a=11 \text{ and } 2b=-3$

$a = 5\frac{1}{2} \quad b = -1\frac{1}{2} \quad (4)$



b) (i) Eqn is of the form

$\left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right) - 1 = 0 \quad -1$

$1 + 2x^2 - x^3 = 0$

or $x^3 - 2x^2 - 1 = 0 \quad -1 \quad (2)$

(ii) Eqn is of the form

$(\sqrt{x})^3 + 2\sqrt{x} - 1 = 0 \quad -1$

$\sqrt{x}(x+2) = 1$

Sq. both sides

$x(x+2)^2 = 1$

$x(x^2 + 4x + 4) = 1$

$x^3 + 4x^2 + 4x - 1 = 0 \quad -1 \quad (2)$

c) (i) $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$

$P'(x) = 4x^3 - 9x^2 - 12x + 28$

$P''(x) = 12x^2 - 18x - 12$

$= 6(2x^2 - 3x - 2)$

$= 6(2x+1)(x-2) \quad -1$

$= 0 \text{ when } x = -\frac{1}{2} \text{ or } 2$

$P\left(-\frac{1}{2}\right) = \frac{1}{16} + \frac{3}{8} - \frac{3}{2} - 14 - 24 = -39\frac{1}{16} \neq 0$

$P(2) = 16 - 24 - 24 + 56 - 24 = 0 \quad -1 \quad (2)$

$\therefore x=2$ is a root of multiplicity 3

(ii) $P(x) = (x-2)^3(ax+b) \quad -1$

Coeff of $x^4 = 1 \therefore a = 1$

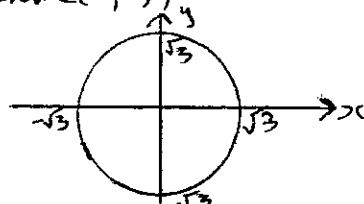
Const. term $= 24 \therefore -8b = 24$

$b = -3 \quad \checkmark$

$\therefore P(x) = (x-2)^3(x+3) \therefore x=2, -3 \quad (2)$

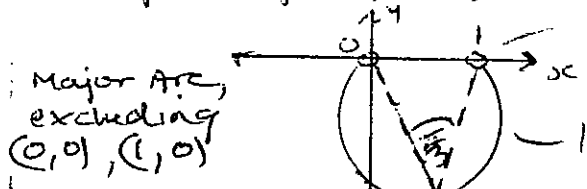
d) (i) $z\bar{z} = 3 \therefore x^2 + y^2 = 3$

Circle, centre (0,0), radius $\sqrt{3}$



(1)

(ii) $\text{Arg } z - \text{Arg}(z-1) = \frac{\pi}{3}$



Major Arc, excluding (0,0), (1,0)

(2)

Q5(a) (i) $(x-y)^2 > 0$ for all x, y
 $x^2 - 2xy + y^2 \geq 0$

$$x^2 + y^2 \geq 2xy \quad -$$

Also $y^2 + z^2 \geq 2yz$
 $y^2 + x^2 \geq 2zx \quad -$

Adding: $2(x^2 + y^2 + z^2) \geq 2(xy + yz + zx)$

$$\therefore x^2 + y^2 + z^2 \geq xy + yz + zx$$

(ii) $x^3 + y^3 + z^3 - 3xyz = (x+y+z)$
 $x(x^2 + y^2 + z^2 - xy - yz - zx)$

$$\geq (x+y+z)(xy + yz + zx - xy - yz - zx)$$

(from part (i)) $-$

$$\geq (x+y+z) \cdot 0$$

$$\geq 0$$

$$\therefore x^3 + y^3 + z^3 \geq 3xyz$$

(iii) In (ii)

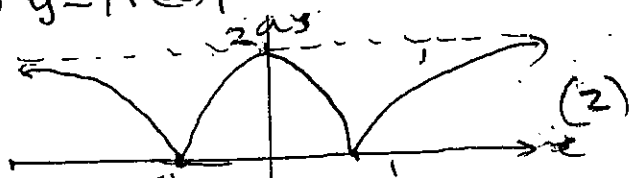
let $x^3 = a, y^3 = b, z^3 = c$

$$\therefore a + b + c \geq 3a^{\frac{1}{3}}b^{\frac{1}{3}}c^{\frac{1}{3}} \quad -$$

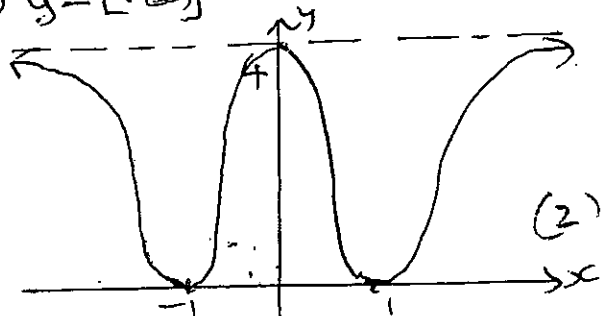
$$\geq 3(abc)^{\frac{1}{3}}$$

(4)

b) (i) $y = |f(x)|$

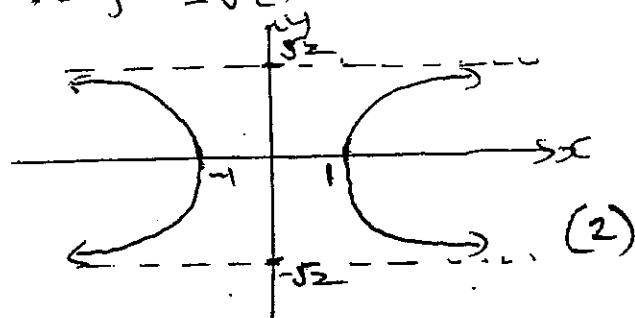


(ii) $y = [f(x)]^2$

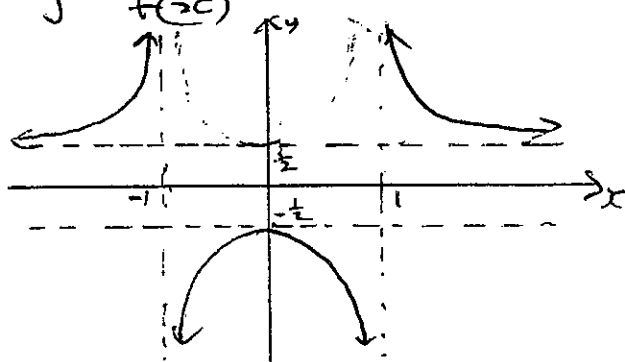


(ii) $y^2 = f(x)$

$$\therefore y = \pm \sqrt{f(x)}$$



(iv) $y = \frac{1}{f(x)}$



c) (i) $x^3 + 3x^2y - 2y^3 = 16$

Diff. imp.

$$3x^2 + 6xy + 3x^2y' - 6y^2y' = 0$$

$$y'(6y^2 - 3x^2) = 3x^2 + 6xy$$

$$\therefore y' = \frac{3(x^2 + 2xy)}{3(2y^2 - x^2)}$$

$$= \frac{x^2 + 2xy}{2y^2 - x^2} \quad (1)$$

(ii) St. pts when $y' = 0$

$$x^2 + 2xy = 0$$

$$x(x + 2y) = 0$$

$$\therefore x = 0, y = -2$$

or $x = -2y$

$$\therefore (-2y)^3 + 3(-2y)^2y - 2y^3 = 16$$

$$-8y^3 + 12y^3 - 2y^3 = 16$$

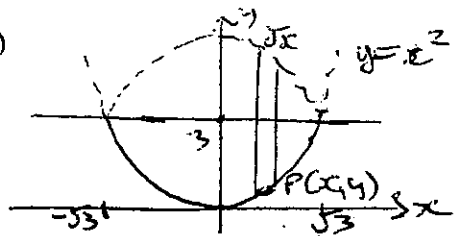
$$2y^3 = 16$$

$$y = 2$$

$$\therefore x = -4$$

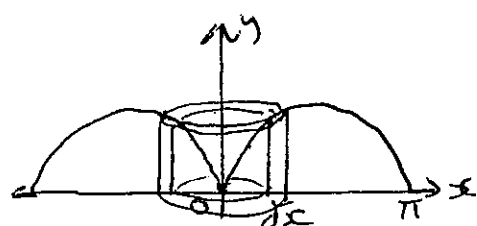
$$\therefore \text{St. pts are } (-4, 2) \text{ and } (0, -2) \quad (2)$$

Q6.(a)
15



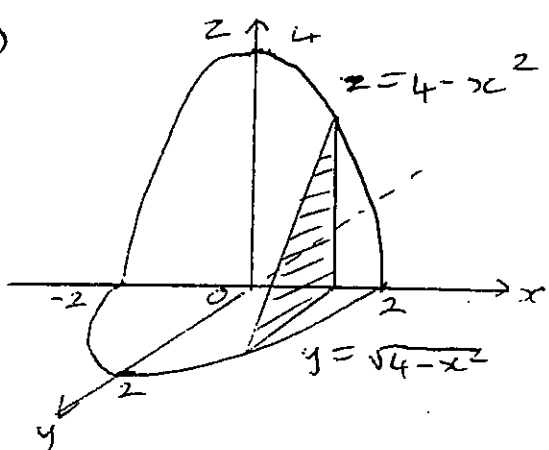
Vol. of slice $\Delta V = \pi r^2 \Delta x$
 $= \pi (3 - y)^2 \Delta x$
 $= \pi (3 - x^2)^2 \Delta x$ — |
 Total Vol., $V = \lim_{\Delta x \rightarrow 0} \sum_{x=-\sqrt{3}}^{\sqrt{3}} \pi (3 - x^2)^2 \Delta x$
 $= 2\pi \int_0^{\sqrt{3}} (3 - x^2)^2 dx$ — |
 $= 2\pi \int_0^{\sqrt{3}} (9 - 6x^2 + x^4) dx$
 $= 2\pi \left[9x - 2x^3 + \frac{x^5}{5} \right]_0^{\sqrt{3}}$
 $= 2\pi \left[9\sqrt{3} - 6\sqrt{3} + \frac{9\sqrt{3}}{5} \right]$
 $= \frac{2\pi}{5} [45\sqrt{3} - 30\sqrt{3} + 9\sqrt{3}]$
 $= \frac{48\pi\sqrt{3}}{5} u^3$ — | (4)

b)



Vol shell = in. circ. $\times$ height $\times$ thick.
 $= 2\pi x y \Delta x$ — |
 Vol. = $\lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi} 2\pi x y \Delta x$
 $= 2\pi \int_0^{\pi} x \sin x dx$ — |
 $= 2\pi [uv - \int v u']$
 $= 2\pi \left[-x \cos x - \int_0^{\pi} \cos x dx \right]$ — |
 $= 2\pi \left[-\pi \cos \pi + [\sin x]_0^{\pi} \right]$
 $= 2\pi [\pi - 0]$
 $= 2\pi^2 u^3$ — | (4)

→ (ii)



(ii) Vol slice = $(\frac{1}{2} b h) H$
 $= \frac{1}{2} y z \Delta x$ — |
 $= \frac{1}{2} \sqrt{4 - x^2} (4 - x^2) \Delta x$
 $= \frac{1}{2} (4 - x^2)^{\frac{3}{2}} \Delta x$ — |
 Vol. = $\lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 \frac{1}{2} (4 - x^2)^{\frac{3}{2}} \Delta x$
 $= 2 \int_0^2 \frac{1}{2} (4 - x^2)^{\frac{3}{2}} dx$ — |
 $= \int_0^2 (4 - x^2)^{\frac{3}{2}} dx$ (3)

(iii) $V = \int_0^2 (4 - x^2)^{\frac{3}{2}} dx$
 let $x = 2 \sin \theta \therefore dx = 2 \cos \theta d\theta$
 when $x = 0$, $\theta = 0$
 when $x = 2$, $\theta = \frac{\pi}{2}$ — |
 $\therefore V = \int_0^{\frac{\pi}{2}} (4 - 4 \sin^2 \theta)^{\frac{3}{2}} \cdot 2 \cos \theta d\theta$
 $= \int_0^{\frac{\pi}{2}} 8 (1 - \sin^2 \theta)^{\frac{3}{2}} \cdot 2 \cos \theta d\theta$
 $= 16 \int_0^{\frac{\pi}{2}} \cos^3 \theta \cdot \cos \theta d\theta$
 $= 16 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta$ — |
 $= 16 \int_0^{\frac{\pi}{2}} \left[\frac{1}{2} (1 + \cos 2\theta) \right]^2 d\theta$
 $= 4 \int_0^{\frac{\pi}{2}} 1 + 2 \cos 2\theta + \cos^2 2\theta d\theta$
 $= 4 \left[\theta + \sin 2\theta \right]_0^{\frac{\pi}{2}} + 4 \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 4\theta) d\theta$
 $= 4 \times \frac{\pi}{2} + 2 \left[\theta + \frac{1}{4} \sin 4\theta \right]_0^{\frac{\pi}{2}}$
 $= 2\pi + \pi - 0$
 $= 3\pi u^3$ — | (3)

Q 7. (a) Let $\frac{6x+4}{(x^2+1)(x+1)} = \frac{ax+b}{x^2+1} + \frac{c}{x+1}$
 $\therefore 6x+4 = (ax+b)(x+1) + c(x^2+1)$
 If $x = -1$, $-2 = c \times 2 \therefore c = -1$
 If $x = 0$, $4 = b + c \therefore b = 5$
 Eq. coeff of x^2
 $a + c = 0 \therefore a = 1$

$\therefore I = \int_0^1 \frac{x+5}{x^2+1} - \frac{1}{x+1} dx$
 $= \int_0^1 \frac{x}{x^2+1} dx + \int_0^1 \frac{5}{x^2+1} dx - \int_0^1 \frac{dx}{x+1}$
 $= \frac{1}{2} [\ln(x^2+1)]_0^1 + 5 [\tan^{-1}x]_0^1 - [\ln(x+1)]_0^1$
 $= \frac{1}{2} (\ln 2 - \ln 1) + 5 (\tan^{-1}1 - \tan^{-1}0) - (\ln 2 - \ln 1)$
 $= \frac{1}{2} \ln 2 - \ln 2 + 5 \times \frac{\pi}{4}$
 $= \frac{5\pi}{4} - \frac{1}{2} \ln 2$ (4)

b) (i) $P(z) = z^4 + bz^2 + d$
 $P(z) = 4z^3 + 2bz$ -
 $P'(z) = (-z)^3 + zb(-z)$
 $= -4z^3 - 2bz$
 $= -(4z^3 + 2bz)$
 $= -P'(z)$ -
 $\therefore P'(z)$ is odd. (2)

ii) $P(-\alpha) = (-\alpha)^4 + b(-\alpha)^2 + d$
 $= \alpha^4 + b\alpha^2 + d$
 $= P(\alpha)$
 $= 0$ -
 $\therefore -\alpha$ is also a zero of $P(z)$
 Let β = the fourth root.
 $\Sigma x = \alpha + \alpha + (-\alpha) + \beta = -\frac{b}{a} = 0$
 $\therefore \beta = -\alpha$ -
 $\therefore -\alpha$ is a double root of $P(z)$ (2)

(c) Result. Force, $ma = mg - mkv$
 $35a = 350 - 70v$ -
 $a = 10 - 2v$
 $\therefore \frac{dv}{dt} = 10 - 2v$
 $t = -\frac{1}{2} \ln(10-2v) + c$ -
 $2t = -\ln(10-2v) + 2c$
 $\ln(10-2v) = 2c - 2t$
 $10-2v = e^{2c-2t} = Ae^{-2t}$
 $2v = 10 - Ae^{-2t}$
 $v = 5 - \frac{A}{2} e^{-2t}$ -
 When $t = 0$, $v = 30$
 $\therefore 30 = 5 - \frac{A}{2}$
 $A = -50$
 $\therefore v = 5 + 25e^{-2t}$ (3)

(ii) As $t \rightarrow \infty$, $e^{-2t} \rightarrow 0$
 $\therefore v \rightarrow 5 \text{ ms}^{-1}$ (1)

(iii) $\frac{dx}{dt} = 5 + 25e^{-2t}$
 $x = 5t - \frac{25}{2} e^{-2t} + c$ -
 When $t = 0$, $x = 0 \therefore c = \frac{25}{2}$
 $\therefore x = 5t - \frac{25}{2} e^{-2t} + \frac{25}{2}$
 When $t = 60$
 $x = 300 - \frac{25}{2} e^{-120} + \frac{25}{2}$
 $= 312\frac{1}{2} - \frac{25}{2} e^{-120}$ -
 ≈ 312.5
 $\therefore \text{Height} = 400 - 312.5$ -
 $= 87.5 \text{ m}$ (3)

28(a) (i) $I_n = \int_0^1 x^n e^{-x} dx$

$I_0 = \int_0^1 x^0 e^{-x} dx$
 $= \int_0^1 e^{-x} dx$ — 1
 $= [e^{-x}]_0^1$
 $= -e^{-1} - e^0$
 $= 1 - e^{-1}$ — 1 (2)

(ii) $I_n = \int_0^1 x^n e^{-x} dx$

Let $u = x^n$, $u' = nx^{n-1}$

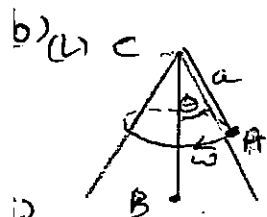
Let $v' = e^{-x}$, $v = -e^{-x}$ — 1

$I_n = uv - \int v u'$
 $= [-x^n e^{-x}]_0^1 + n \int_0^1 x^{n-1} e^{-x} dx$
 $= -e^{-1} + n I_{n-1}$

$I_n = n I_{n-1} - \frac{1}{e}$ (2)

(iii) $I_3 = \int_0^1 x^3 e^{-x} dx$

$= 3 I_2 - \frac{1}{e}$
 $= 3 [2 I_1 - \frac{1}{e}] - \frac{1}{e}$ — 1
 $= 6 I_1 - \frac{4}{e}$
 $= 6 [I_0 - \frac{1}{e}] - \frac{4}{e}$
 $= 6 I_0 - \frac{10}{e}$
 $= 6 [1 - e^{-1}] - \frac{10}{e}$
 $= 6 - \frac{16}{e}$ — 1 (2)



(1) $mg = T \cos \theta + R \sin \theta$ — (1)

$mg = T \cos \theta + R \sin \theta$ — (2)

$mg = T$ — 1

$\therefore$ In (2) $mg = mg \cos \theta + R \sin \theta$

$R \sin \theta = mg(1 - \cos \theta)$

$R = \frac{mg(1 - \cos \theta)}{\sin \theta}$ — 1

In (1) $m a \sin \theta \omega^2 = mg \sin \theta - R \cos \theta$

$m a \omega^2 \sin \theta = mg \sin \theta - \frac{mg(1 - \cos \theta)}{\sin \theta} \cos \theta$

$a \omega^2 \sin^2 \theta = g \sin^2 \theta - g(1 - \cos \theta) \cos \theta$

$\sin^2 \theta (g - a \omega^2) = g \cos \theta (1 - \cos \theta)$

$g - a \omega^2 = \frac{g \cos \theta (1 - \cos \theta)}{(\sin \theta)^2}$

$a \omega^2 = g - \frac{g \cos \theta}{1 + \cos \theta}$

$= \frac{g + g \cos \theta - g \cos \theta}{1 + \cos \theta} \Rightarrow \omega^2 = \frac{g}{1 + \cos \theta}$

(iii) When $R > 0$

$\frac{mg(1 - \cos \theta)}{\sin \theta} > 0$

$1 - \cos \theta > 0$

$\cos \theta < 1$

$\therefore 0^\circ < \theta < 90^\circ$

When $\theta = 0^\circ$, $\omega^2 = \frac{g}{a(1+1)} = \frac{g}{2a}$

$\therefore a = \frac{g}{2\omega^2}$ — 1

When $\theta = 90^\circ$, $\omega^2 = \frac{g}{a(1+0)} = \frac{g}{a}$

$\therefore a = \frac{g}{\omega^2}$ — 1

$\therefore \frac{g}{2\omega^2} < a < \frac{g}{\omega^2}$ (2)

c) $\frac{d}{dx} (x - \frac{1}{3}x^3) = 1 - x^2$

$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$

$\frac{d}{dx} (x - \frac{1}{3}x^3 + \frac{1}{3}x^5) = 1 - x^2 + \frac{5}{3}x^4$ — 1

Now, $\frac{1}{1+x^2} - (1 - x^2) = \frac{1 - (1 - x^4)}{1+x^2} = \frac{x^4}{1+x^2}$

> 0 for $x > 0$

$\therefore 1 - x^2 < \frac{1}{1+x^2}$ — 1

Also

$1 - x^2 + \frac{5}{3}x^4 - \frac{1}{1+x^2} = \frac{(1+x^2)(1 - x^2 + \frac{5}{3}x^4) - 1}{1+x^2}$

$= \frac{1 - x^2 + \frac{5}{3}x^4 + x^2 - x^4 + \frac{5}{3}x^6 - 1}{1+x^2}$

$= \frac{\frac{2}{3}x^4 + \frac{5}{3}x^6}{1+x^2} > 0$ for $x > 0$

$\therefore \frac{1}{1+x^2} < 1 - x^2 + \frac{5}{3}x^4$ — 1

$\therefore 1 - x^2 < \frac{1}{1+x^2} < 1 - x^2 + \frac{5}{3}x^4$

Integrating each exp. gives:

$\int_0^x (1 - x^2) dx < \int_0^x \frac{1}{1+x^2} dx < \int_0^x (1 - x^2 + \frac{5}{3}x^4) dx$

$[x - \frac{x^3}{3}]_0^x < [\tan^{-1} x]_0^x < [x - \frac{1}{3}x^3 + \frac{1}{3}x^5]_0^x$

i.e.

$x - \frac{x^3}{3} < \tan^{-1} x < x - \frac{1}{3}x^3 + \frac{1}{3}x^5$ (4)